Soil Lab Analysis Procedure
Supporting Document `IndigoCarbon_US-1_2023_0015`

June 28, 2024

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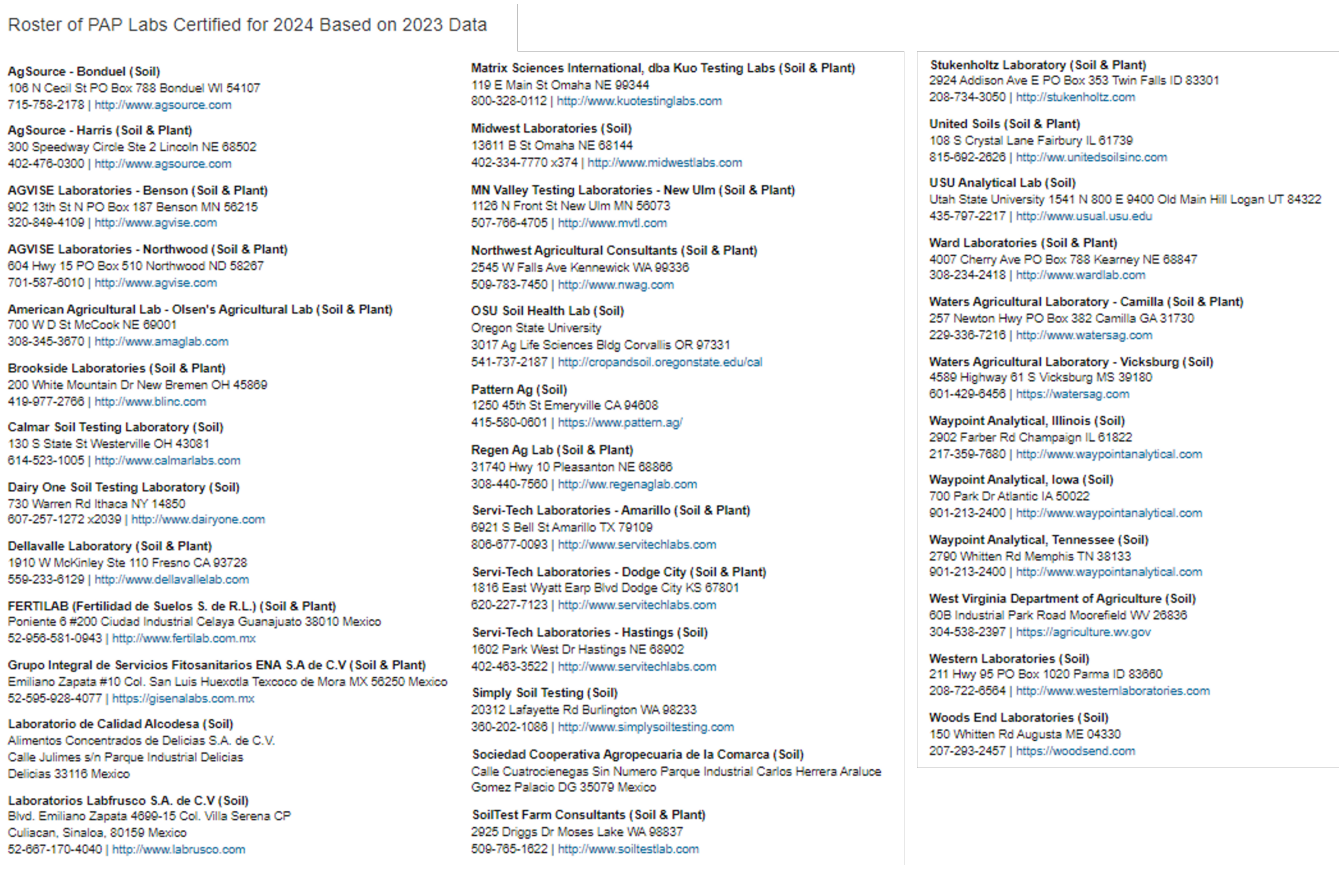
1 Context

The precision and accuracy of the soil carbon measurements quantified via dry combustion, using an elemental analyzer, from third party laboratory partners are monitored using check samples, duplicate measurements, and systematic reanalysis of outlier samples. This page outlines the QC criteria that are used to ensure measurement fidelity.

2 Soil Sample Analysis

In addition to the following processes, laboratories submitting analysis results for Indigo Carbon must participate in the Performance Assessment Program (PAP) offered by the North American Proficiency Testing Program (NAPTP). Members are not provided with a formal accreditation document; however, the annual list of participants can be reviewed on the program website (https://www.naptprogram.org/pap). Figure 1 presents the list as of May 28, 2024.

To our knowledge, state-level accreditation for the soil laboratories contributing to Indigo Carbon is not available.



Roster of PAP Labs Certified for 2024 Based on 2023 Data

AgSource - Bonduel (Soil)
106 N Cecil St PO Box 788 Bonduel WI 54107
715-758-2178 | http://www.agsource.com

AgSource - Harris (Soil & Plant)
300 Speedway Circle Ste 2 Lincoln NE 68502
402-476-0300 | http://www.agsource.com

AGVISE Laboratories - Benson (Soil & Plant)
902 13th St N PO Box 187 Benson MN 56215
320-849-4109 | http://www.agvise.com

AGVISE Laboratories - Northwood (Soil & Plant)
604 Hwy 15 PO Box 510 Northwood ND 58267
701-587-6010 | http://www.agvise.com

American Agricultural Lab - Olsen's Agricultural Lab (Soil & Plant)
700 W D St McCook NE 69001
308-345-3670 | http://www.amaglab.com

Brookside Laboratories (Soil & Plant)
200 White Mountain Dr New Bremen OH 45869
419-977-2766 | http://www.blinc.com

Calmar Soil Testing Laboratory (Soil)
130 S State St Westerville OH 43081
614-523-1005 | http://www.calmarlabs.com

Dairy One Soil Testing Laboratory (Soil)
730 Warren Rd Ithaca NY 14850
607-257-1272 x2039 | http://www.dairyone.com

Dellavalle Laboratory (Soil & Plant)
1910 W McKinley Ste 110 Fresno CA 93728
559-233-6129 | http://www.dellavallelab.com

FERTILAB (Fertilidad de Suelos S. de R.L.) (Soil & Plant)
Poniente 6 #200 Ciudad Industrial Celaya Guanajuato 38010 Mexico
52-956-581-0943 | http://www.fertilab.com.mx

Grupo Integral de Servicios Fitosanitarios ENA S.A de C.V (Soil & Plant)
Emiliano Zapata #10 Col. San Luis Huexotla Texcoco de Mora MX 56250 Mexico
52-595-928-4077 | https://gisenalabs.com.mx

Laboratorio de Calidad Alcodesa (Soil)
Alimentos Concentrados de Delicias S.A. de C.V.
Calle Julimes s/n Parque Industrial Delicias
Delicias 33116 Mexico

Laboratorios Labfrusco S.A. de C.V (Soil)
Blvd. Emiliano Zapata 4699-15 Col. Villa Serena CP
Culiacan, Sinaloa, 80159 Mexico
52-667-170-4040 | http://www.labrusco.com

Matrix Sciences International, dba Kuo Testing Labs (Soil & Plant)
119 E Main St Omaha NE 99344
800-328-0112 | http://www.kuotestinglabs.com

Midwest Laboratories (Soil)
13611 B St Omaha NE 68144
402-334-7770 x374 | http://www.midwestlabs.com

MN Valley Testing Laboratories - New Ulm (Soil & Plant)
1126 N Front St New Ulm MN 56073
507-766-4705 | http://www.mvtl.com

Northwest Agricultural Consultants (Soil & Plant)
2545 W Falls Ave Kennewick WA 99336
509-783-7450 | http://www.nwag.com

OSU Soil Health Lab (Soil)
Oregon State University
3017 Ag Life Sciences Bldg Corvallis OR 97331
541-737-2187 | http://cropandsoil.oregonstate.edu/cal

Pattern Ag (Soil)
1250 45th St Emeryville CA 94608
415-580-0601 | https://www.pattern.ag/

Regen Ag Lab (Soil & Plant)
31740 Hwy 10 Pleasanton NE 68866
308-440-7560 | http://ww.regenaglab.com

Servi-Tech Laboratories - Amarillo (Soil & Plant)
6921 S Bell St Amarillo TX 79109
806-677-0093 | http://www.servitechlabs.com

Servi-Tech Laboratories - Dodge City (Soil & Plant)
1816 East Wyatt Earp Blvd Dodge City KS 67801
620-227-7123 | http://www.servitechlabs.com

Servi-Tech Laboratories - Hastings (Soil)
1602 Park West Dr Hastings NE 68902
402-463-3522 | http://www.servitechlabs.com

Simply Soil Testing (Soil)
20312 Lafayette Rd Burlington WA 98233
360-202-1086 | http://www.simplysoiltesting.com

Sociedad Cooperativa Agropecuaria de la Comarca (Soil)
Calle Cuatrocienegas Sin Numero Parque Industrial Carlos Herrera Araluce
Gomez Palacio DG 35079 Mexico

SoilTest Farm Consultants (Soil & Plant)
2925 Driggs Dr Moses Lake WA 98837
509-765-1622 | http://www.soiltestlab.com

Stukenholtz Laboratory (Soil & Plant)
2924 Addison Ave E PO Box 353 Twin Falls ID 83301
208-734-3050 | http://stukenholtz.com

United Soils (Soil & Plant)
108 S Crystal Lane Fairbury IL 61739
815-692-2626 | http://ww.unitedsoilsinc.com

USU Analytical Lab (Soil)
Utah State University 1541 N 800 E 9400 Old Main Hill Logan UT 84322
435-797-2217 | http://www.usual.usu.edu

Ward Laboratories (Soil & Plant)
4007 Cherry Ave PO Box 788 Kearney NE 68847
308-234-2418 | http://www.wardlab.com

Waters Agricultural Laboratory - Camilla (Soil & Plant)
257 Newton Hwy PO Box 382 Camilla GA 31730
229-336-7216 | http://www.watersag.com

Waters Agricultural Laboratory - Vicksburg (Soil)
4589 Highway 61 S Vicksburg MS 39180
601-429-8456 | https://watersag.com

Waypoint Analytical, Illinois (Soil)
2902 Farber Rd Champaign IL 61822
217-359-7680 | http://www.waypointanalytical.com

Waypoint Analytical, Iowa (Soil)
700 Park Dr Atlantic IA 50022
901-213-2400 | http://www.waypointanalytical.com

Waypoint Analytical, Tennessee (Soil)
2790 Whitten Rd Memphis TN 38133
901-213-2400 | http://www.waypointanalytical.com

West Virginia Department of Agriculture (Soil)
60B Industrial Park Road Moorefield WV 26836
304-538-2397 | https://agriculture.wv.gov

Western Laboratories (Soil)
211 Hwy 95 PO Box 1020 Parma ID 83660
208-722-6564 | http://www.westernlaboratories.com

Woods End Laboratories (Soil)
150 Whitten Rd Augusta ME 04330
207-293-2457 | https://woodsend.com

Figure 1: 2024 NAPTP – PAP accredited laboratory list from the program website.

3 Carbon Samples

3.0.1 Sample Preparation

- All samples require preliminary treatment consisting of drying to constant weight, grinding, and homogenization (i.e., well-mixed). If the goal of the assessment includes knowing how much inorganic carbon the soil contains, pre-treating a subset of the sample units with a strong acid is necessary.
  - All samples should be air-dried within 48 hours of arriving at the laboratory facility or stored under refrigeration until they can be dried.
  - Dried samples should be ground and sieved to 2 mm using a process that will disaggregate large clumps of soil, but will not pulverize rock.
  - Samples sieved to < 2 mm fraction. The coarse fraction (> 2 mm) is discarded.

3.0.2 Sample Analysis

- The dry combustion method using an elemental analyzer is the recommended method to estimate soil carbon content in all kinds of soils for its accuracy and efficiency.
- Dry combustion is conducted by heating (~1000 °C) a soil-catalyst mixture in a resistance furnace or induction furnace in a stream of O_2 or CO_2-free air, followed by quantitation of evolved CO_2.
- All prepared carbon samples are tested for excess lime using an acid drop "fizz test" procedure prior to measurement and then organic carbon is quantified using the 'dry combustion' method with an elemental analyzer.
  - If the test is positive (visible fizzing occurs when the sample is exposed to acid indicating the presence of free carbonates), the samples should be analyzed for total organic carbon using the laboratory's typical procedure in which inorganic carbon is removed prior to analysis:
    - Sample pretreatment includes small applications of either dilute hydrochloric or sulfurous acid to the sample until visible effervescence ceases, followed by drying of the sample prior to analysis.
- If the test is negative (no visible fizzing occurs when the sample is exposed to acid indicating the absence of free carbonates), samples should be analyzed for total carbon using the laboratory's typical procedure. The negative excess lime test indicates that it is safe to assume that total carbon is equivalent to total organic carbon.
- All dry combustion carbon results will be reported as percent carbon (% C) by mass.
- Following analysis, all samples are retained for 4 weeks.

4 Bulk Density Samples

4.0.1 Sample Preparation

- All samples should be dried within 48 hours of arriving at the lab's facility or stored under refrigeration until they can be dried.
- Dried samples should be ground and sieved to 2 mm using a process that will disaggregate large clumps of soil but will not pulverize rock.

4.0.2 Sample Analysis

- Samples will be dried at 105°C to remove all water content.
- Dry mass of the fine (< 2 mm) fraction will be reported in grams.
- If samples contain $> 5\%$ rock content by visual inspection, the mass of the coarse (> 2 mm) and fine material will both be reported separately in grams.
- Following analysis, all samples should be retained for 4 weeks.

4.0.3 Bulk Density Calculation

- To ensure that the appropriate volume is used in the calculation of bulk density, the raw sample mass reported by the labs will be converted to bulk density by Indigo.
- BD (g/cm^3) = fine earth fraction mass in g / (3.14 x (2.54 cm x 2.54 cm) x sample depth in cm).
- For samples where the full 30.48 cm depth was reached:
  - BD (g/cm^3) = fine earth fraction mass in g / (617.46 cm^3)
- NOTE: Coarse material is not included in the bulk density calculation as this fraction is not known to contain significant amounts of carbon, nor is this fraction represented in the carbon analysis.

5 Texture and pH Samples

5.0.1 Sample Preparation

- All samples should be dried within 48 hours of arriving at the contractor's facility or stored under refrigeration until they can be dried.
- Dried samples should be ground and sieved to 2 mm using a process that will disaggregate large clumps of soil, but will not pulverize rock.
- Analysis should be performed only on the fine < 2 mm fraction.

5.0.2 Sample Analysis

- Texture analysis should be performed using the hydrometer method according to the contractor's typical laboratory procedure. Results should be reported as percent silt, percent sand, and percent clay.
- pH should be measured using a standard hydrogen electrode (SHE) on a 1:1 mixture of water and soil according to the contractor's typical procedure.
- Following analysis, all samples should be retained for 4 weeks.

6 Soil Lab QC Process

6.0.1 Soil Organic Carbon Check Sample Analysis

A check sample analysis is used to measure carbon measurement accuracy from analysis batch to analysis batch. At least one check sample is reported for each soil carbon analysis date. If the reported check sample value in a batch is below 92% or above 108% of the certified or standard check sample value provided to Indigo prior to measurement, the full analytical run is reanalyzed.

6.0.2 Soil Organic Carbon Duplicate Sample Analysis

A duplicate sample analysis on a subset of samples was used to measure within-batch precision and instrument drift within analytical runs.

For samples measuring $<$ 2.5 % SOC

- A duplicate is considered ACCEPTABLE if the absolute difference between the two duplicate samples is $\leq$ 0.2 % SOC.
- If the absolute difference between the two duplicate samples is $>$ 0.2 % SOC, the duplicate set is considered UNACCEPTABLE.

For samples measuring $\geq$ 2.5 % SOC

- A duplicate is considered ACCEPTABLE if the absolute difference between the two duplicate samples is $\leq$ 8% of the first measured value.
- If the absolute difference between the two duplicate samples $>$ 8% of the first measured value, the duplicate set is considered UNACCEPTABLE.

Resolution of UNACCEPTABLE sample measurements If only one set of duplicate samples for a reported analytical run is UNACCEPTABLE, the original duplicate sample along with the two samples measured immediately preceding, and two samples immediately following the duplicate set according to lab sample id number (5 total samples) are reanalyzed.

If the absolute or relative difference of all reanalysis results between the original reported measurement (sample not labelled as duplicate in the case of the original duplicate set) and reanalyzed samples meet the requirements outlined as ACCEPTABLE (as defined above), the samples are accepted and the batch passes. If the absolute or relative difference of any reanalyzed pair is UNACCEPTABLE (as defined above), the full analytical run is reanalyzed.

6.0.3 Soil Organic Carbon Outlier Sample Analysis (03-01-2021 - Present)

Reanalysis (for confirmation purposes) of abnormally high or low measurements that are part of analytical batches that fully pass both check and duplicate sample procedures is required from time to time.

Minimum Sample Selection (more may be selected on a discretionary basis)

SOC samples requiring confirmation analysis are selected from fields when all expected results are returned from the lab and the sample Z-Score $\geq$ 2.5 or $\leq$ -2.5.

Z-Score Calculation:

(sample SOC measurement - call plan mean) / (call plan standard deviation)

call plan = all samples collected on the respective field within a single sampling season

Reanalysis Results Assessment

Reanalysis results are assessed according to the rules outlined as acceptable for duplicate measurements in the "Acceptable Duplicate Measurement" section. If the reanalysis of more than two samples in an analytical batch is found to exceed the acceptable threshold, the entire batch fails and will be reanalyzed at no cost to Indigo.

- Analytical Batch:
  - 1 check sample for analysis date: Samples reported within a single submitted file.
  - $>$ 1 check sample for analysis date: Samples between check samples according to lab sample id reported in a single file (corresponds to a single instrument run).

Determination of Accepted Result

Samples that fail reanalysis require a third analysis (second reanalysis) to confirm the correct sample SOC value. The following criteria determine which result value is accepted as correct:

- If the 1st reanalysis passes, then the original analysis result is accepted.
- If the 1st reanalysis fails but the second reanalysis (third analysis) passes with respect to the original analysis, then the original analysis result is accepted.
- If the 1st reanalysis fails and the second reanalysis (third analysis) passes with respect to the 1st reanalysis, then the 1st reanalysis result is accepted.
- If none of the samples pass the above criteria, the sample is discarded.
- If reanalysis is not performed on a selected sample for any reason, the sample is discarded.

- Should correction/deletion of a sample according to this procedure cause a new, previously unflagged sample to be flagged, the newly flagged sample may be, but is not required to be, reanalyzed.

6.0.4 Bulk Density

- The only action performed by the Contractors for bulk density analysis beyond sample preparation is the recording of sample mass. As a result, no duplicate or check samples are used to monitor analytical performance. It is trusted to internal laboratory QA/QC processes to keep balances for recording sample mass in working order.

6.0.5 Texture and pH

- Each Contractor employs internal processes to ensure pH and texture analysis QA/QC quality and is accredited by the NAPTP-PAP program. Given the relative insensitivity of the modelling output to these parameters, this is deemed sufficient for Indigo's purposes and does not pose a risk to measurement integrity.

7 Summary of Changes

Section	Date	Summary of Changes
Soil Sample Analysis	May 28th, 2024	Updated Figure 1